NVIDIA

Toward Real-Time Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

Shunyang Li | Benchmarks measured | 27 Oct 2026



Agenda

01 Toward Real-Time H3 Generation

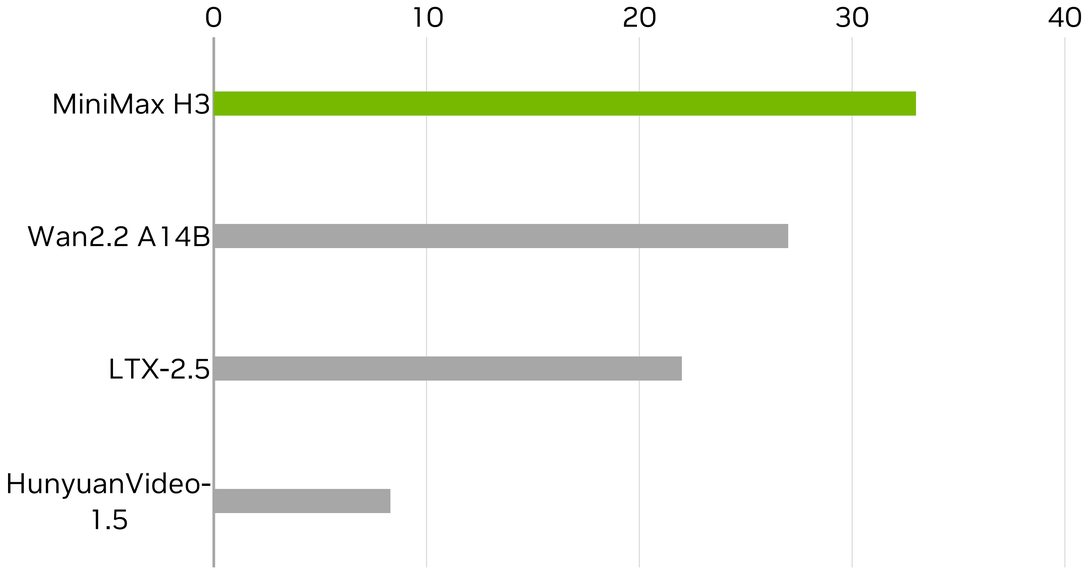
02 vLLM-Omni and ComfyUI Applications



01 · Real-Time H3 Generation

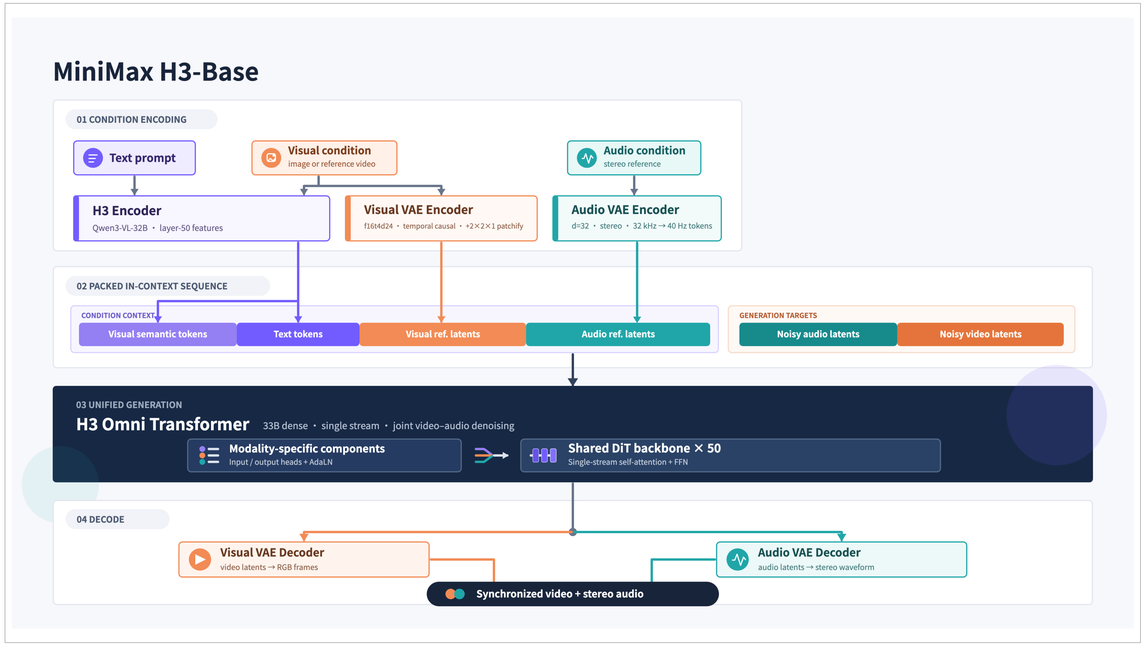
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

A 32B encoder feeds the 33B Omni Transformer and visual/audio VAEs. Fifty transformer blocks are evaluated at each of 50 baseline steps.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

NVIDIA

SM120 Brings FP4 Compute; PCIe Limits Scale

All discrete SM120 cards pair fifth-generation Tensor Cores with GDDR7 and PCIe Gen5; none provides GPU-to-GPU NVLink

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 ×8 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: RDMA via NIC*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

NVIDIA

Baseline: Configuration and Breakdown

Original Ref2VA · 50 evaluations · BF16 · dense attention · 8 SM120 GPUs

BASELINE LATENCY

213.402 s

Complete-MP4 HTTP median · n = 2

GENERATED VIDEO

960 × 544

124 frames · 24 FPS · 5.167 s

HARDWARE

8 × SM120 GPUs

PRO 5000 reference · 73,415 MiB / GPU

Reference configuration

Original Ref2VA · BF16 DiT · 50 evaluations

Dense cuDNN attention; SDPA token refiner

DiT TP1 / Ulysses8 · text TP8 · VAE tiles8

Local NCCL · native VAE · FP16 decode

Exact AdaLN table resident before timing

Fixed inputs / seed 43; warmup excluded

Request-level latency breakdown

Model engine	**211.899 s**
Output-ready wait	**1.330 s**
MP4 encoding	**0.145 s**
Other HTTP overhead	**0.028 s**

Model engine includes prompt encoding, DiT, VAE and coordination.

Completed requests: 214.705 s and 212.099 s · third request interrupted and excluded

Breakdown = mean of the same two requests, using service logs and HTTP timing. No new Nsight capture.

NVIDIA

Five-Second Avatar: Live Conversation

Demo focus · reference identity, expression, native speech and continuity

01 · Natural presence

Listen with natural presence.
Microphone stays active; live captions show the question.

02 · Speech and expression

Generate a short answer as native audio and video.
Show generation time and actual reply-video start time.

03 · Continuous generation

Carry video and audio history across windows.
Inspect the face, mouth and transitions while speaking.

960 × 544 · 24 FPS · 8 steps · 8 SM120 GPUs · QAD NVFP4 / native FP4 VSA

119 new frames / 4.958 s per continuation · 20% history noise · video prefix off.
ASR and LLM share GPUs 6 / 7. Full configuration and timing definitions are on slide 21.

NVIDIA

Three Ideas Behind Faster Generation

Technical overview · detailed algorithms and source excerpts remain available on hidden slides

Less model work

Few-step distillation reduces evaluations.
Sparse attention concentrates work on selected regions.
Quantized inference lowers arithmetic and memory cost.

Overlap across eight GPUs

Ulysses distributes attention across heads.
RDMA carries the exchanges.
Head bands overlap transfer with attention compute.

Keep the output path moving

CUDA Graph replays stable head-band compute.
Fused packing reduces intermediate copies.
NVFP4 VAE and MP4 pre-encoding reduce output cost.

These are optimization families. Use each demo’s own recipe and measurements; the four-step avatar and slide 18 are distinct workloads.

NVIDIA

Optimization Path: Recorded Latency

960 × 544 · 124 frames / 24 FPS · 8 exclusive GPUs · same inputs and seed 43

Stage	Cumulative change	Median (s)	Saved (s)	RTF
00	Baseline · original BF16 / dense / 50 evaluations	213.402	—	41.304
01	+ PDD · 50 → 8 evaluations	44.327	—	8.579
02	+ Trained VSA · native 8-step model / runtime	25.915	+18.412	5.016
03	+ QAD NVFP4 DiT / native FP4 attention	19.460	+6.455	3.767
04	+ RDMA / direct layout / V-gate overlap	6.832	+12.628	1.322
05	+ 2 / 3 / 2 head pipeline	6.407	+0.425	1.240
06	+ Compute CUDA graphs	6.193	+0.214	1.199
07	+ 36-layer NVFP4 VAE / MP4 pre-encoding	5.420	+0.773	1.049
08	+ Fused tile preparation / early AAC	5.194	+0.226	1.005
09	+ Direct output packing	5.147	+0.047	0.996
10	+ Direct receive-to-tile input	5.060	+0.087	0.979
11	+ Integer routing compaction	5.015	+0.045	0.971
12	**+ Producer Q/K/V copy elision**	**4.922**	**+0.092**	**0.953**

Final 4.922 s · recheck 4.927 s · 6/6 < 5 s · 4.5 s target not reached

Baseline refreshed: n = 2. Later rows / recheck retained: n = 3. First transition has no isolated gain claim.

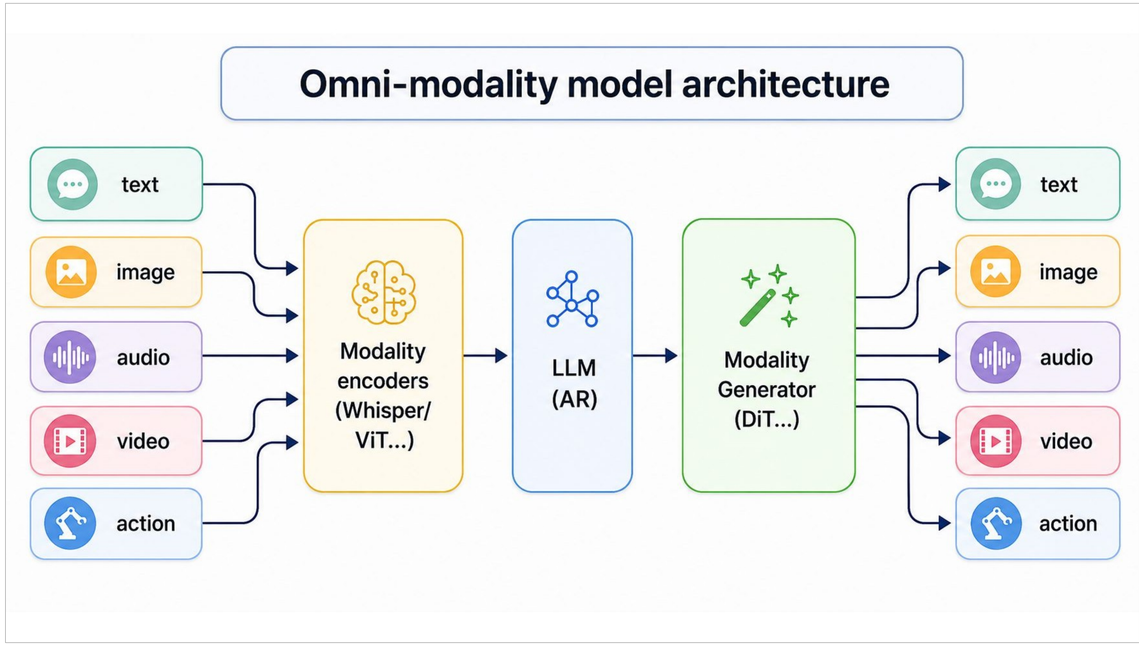
RTF = latency / 5.167 s. Model, schedule and precision changes are not all lossless. ASR / LLM are excluded.

NVIDIA

02 · ComfyUI Apps With vLLM-Omni

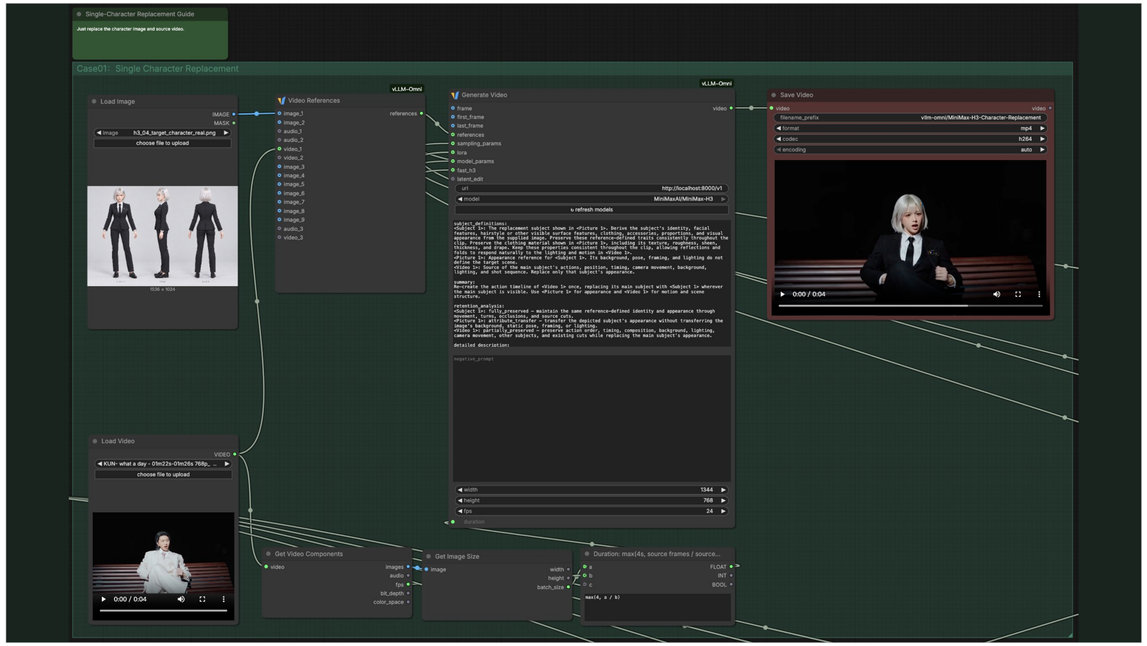
vLLM-Omni Brings Multi-GPU H3 to ComfyUI

PR #9194 adds character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI authors the workflow; vLLM-Omni runs scheduling, model execution, and multi-GPU deployment



ComfyUI Workflow · PR #9194

The workflow supports single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194

NVIDIA

Live Avatar: Configuration and Timing

Microphone → VAD / ASR → LLM → H3 audio + video → browser playback

OUTPUT

960 × 544 · 24 FPS

124-frame first window: 5.167 s

CONTINUATION

119 new frames: 4.958 s

5-frame overlap · video + audio history

HARDWARE

8 × SM120 GPUs

Recorded PRO 5000 node · BF16 RDMA

H3 generation

8 evaluations · QAD NVFP4 DiT · native FP4 VSA
TP1 / Ulysses8 · text TP8 / VAE PP8 · 2/3/2 heads

Decoder / history

36-layer NVFP4 VAE · MP4 pre-encoding
20% video-history noise · target video prefix off

Speech detection / ASR

CPU Silero VAD · 600 ms quiet endpoint
Qwen3-ASR-0.6B · BF16 · GPU 6, shared with H3

LLM / speech output

Qwen3-4B · BF16 · GPU 7, shared with H3
Streaming short replies · native H3 speech, no separate TTS

Two different clocks

Avatar generation

H3 request → complete MP4.
Excludes later audio screening and delivery.

Reply video starts

End of your speech → first reply frame.
Includes VAD, ASR, LLM, queueing, generation and playback wait.

One active call · fullscreen avatar · live captions · microphone + avatar audio recording

Demo shares GPUs 6 / 7 and carries history. Slide 18 is a standalone, fixed-input H3 benchmark with ASR / LLM stopped.

NVIDIA


NVIDIA